Learned bias correction for surface-area estimation from CT-like voxel data

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Abstract

Surface area measured from voxelized CT data is biased: voxel-face counting overestimates it by about 50% at every resolution, marching cubes on binary data does not converge as the grid is refined, and marching cubes on greyscale data is pulled down by curvature and pushed up by noise. We build a synthetic CT-like benchmark of 2,400 shapes from seven families, with reference surface areas that are exact or converged to better than 10^{-4}, rendered as 14,400 volumes at 32^3 and 64^3 with three noise levels. On this benchmark we compare four classical estimators, a per-regime lookup correction, a Random Forest on hand-crafted descriptors, a 3D CNN, a PointNet and a hybrid CNN that also receives the descriptors. All learned models predict a multiplicative correction to the greyscale marching-cubes area. A simple heat-equation argument predicts the curvature bias of greyscale marching cubes on spheres closely (-0.56% predicted against -0.62% measured at 32^3). On held-out test shapes, greyscale marching cubes has a mean absolute error of 4.78% (32^3) and 3.52% (64^3); the best corrections reach 0.62% (hybrid, 32^3) and 0.18% (Random Forest, 64^3), and stay below 2% on two shape families never seen in training. The Random Forest matches the deep models overall, the image-based models are best on objects smaller than 16 voxels, and PointNet is clearly weaker. No method recovers structures thinner than about two voxels.

1 Introduction

Surface area is one of the most requested measurements in X-ray computed tomography (CT) of materials and parts. It controls reaction rates in catalysts, flow and filtration behaviour in porous media, and the quality of additively manufactured lattices. In practice it is measured from a reconstructed voxel volume, either by counting the exposed faces of segmented voxels or by extracting a triangle mesh with marching cubes [1] and summing the triangle areas.

Both routes are biased. Counting voxel faces measures the area of a staircase, which is about 50% larger than the true surface at every resolution. Marching cubes on a thresholded (binary) volume is better, but it is a local method with a finite set of triangle orientations and its bias does not vanish as the voxel size shrinks [2, 3]. Marching cubes on the greyscale volume uses the partial-volume information and converges for clean, well-resolved data, but real CT volumes are noisy and many structures of interest are only a few voxels across. Noise roughens the extracted surface and adds small spurious components, while smoothing the volume to suppress noise shrinks curved surfaces and can erase thin features altogether.

The size and sign of the error therefore depend on resolution, noise, curvature and topology at the same time, which makes a single fixed correction factor inadequate. This is a natural setting for a learned correction: a model that looks at the volume, or at descriptors computed from it, and predicts how far the classical measurement is from the truth for that particular object. Training such a model needs ground truth, which is only available exactly for synthetic objects.

This work builds a synthetic CT-like dataset of seven shape families with reference surface areas that are exact or converged to better than 10^{-4}, renders them at two resolutions and three noise levels, and compares four classical estimators, two calibrated baselines and three learned models (a 3D convolutional network, a PointNet on marching-cubes surface points, and a hybrid network

that combines the volume with hand-crafted descriptors). The models predict a multiplicative correction to the greyscale marching-cubes area, so that learning acts as a bias-correction layer on top of the classical measurement rather than replacing it.

2 Problem, objectives and motivation

Problem. Given a CT-like greyscale volume of a single object on an N^3 grid, estimate the area of the object's true, continuous surface. The volume is degraded by the finite voxel size (partial-volume averaging), by the system point-spread function and by noise.

Objectives.

1. Build a library of 3D shapes with reference surface areas accurate enough (relative error below 10^{-4}) that the errors of all estimators can be measured without contamination from the reference itself.
2. Quantify the bias and spread of the classical estimators (voxel-face counting and three marching-cubes variants) as a function of resolution and noise.
3. Train a 3D CNN and a PointNet to correct the greyscale marching-cubes area, and compare them, at matched resolution and noise, with the classical estimators and with two calibrated baselines: a per-regime lookup table and a Random Forest on hand-crafted descriptors.
4. Test whether the learned corrections transfer to shape families never seen in training (superellipsoids with sharp edges or tips, and blobs with fine surface roughness).
5. State when a learned correction is worth using and when a classical estimator is enough.

Why this matters. A 5% error in surface area is large compared with the effects that surface-area measurements are often used to detect, for example differences between catalyst batches, changes in pore structure, or the deviation of a printed lattice from its design. Errors of this size are typical for marching cubes on binary data and for greyscale marching cubes on small or noisy objects (Section 6). A correction that is learned once on synthetic data and then applied per object is cheap at inference time and can be reused for any scanner whose resolution and noise level fall inside the training range.

3 Theory

3.1 Voxel-face counting

Consider a planar piece of surface with unit normal $\mathbf{n} = (n_x, n_y, n_z)$. Its digital (staircase) version is made of voxel faces normal to the three axes, and the total area of the faces normal to axis i equals the projection of the true area onto the plane perpendicular to that axis. The face-count area per unit true area is therefore

$$r(\mathbf{n}) = |n_x| + |n_y| + |n_z|, \qquad 1 \le r \le \sqrt{3}. \tag{1}$$

It equals 1 only for axis-aligned surfaces (Fig. 1). For normals distributed uniformly on the sphere each component is uniform on $[-1, 1]$, so $\mathbb{E}|n_i| = 1/2$ and $\mathbb{E}\, r = 3/2$. The face count of a randomly oriented smooth object thus overestimates its area by 50%, and since r does not depend on the voxel size, refining the grid does not help. On a clean sphere at 128^3 we measure +50.03% (Table 1).

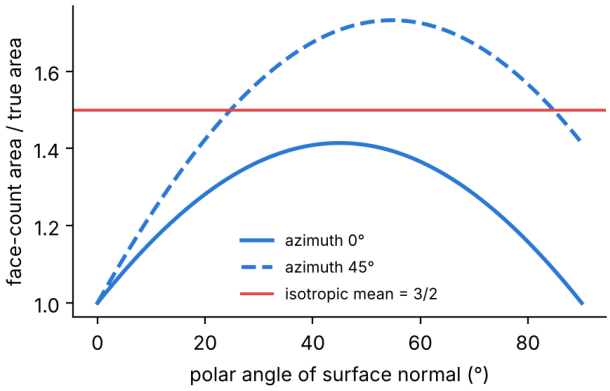


Figure 1: Face-count area per unit true area, $|n_x| + |n_y| + |n_z|$, for a plane whose normal makes the given polar angle with the z axis, at azimuth 0° and 45°. The isotropic average is 3/2.

Table 1: Greyscale marching-cubes error on clean spheres: measured vs the curvature prediction $-2\sigma_{\text{eff}}^2/R^2$ from Eq. (2). R in voxels.

		MC on greyscale		MC after smoothing ($\sigma_s = 1$)	
Grid	R	predicted	measured	predicted	measured
16^3	5.4	−3.04%	−3.65%	−9.90%	−10.99%
32^3	12.6	−0.56%	−0.62%	−1.82%	−2.00%
64^3	27.0	−0.12%	−0.05%	−0.40%	−0.43%
128^3	55.8	−0.03%	+0.07%	−0.09%	−0.10%

3.2 Marching cubes on binary and greyscale volumes

Marching cubes [1] places triangle vertices on the edges of each $2\times2\times2$ cell where the image crosses the iso-level. On a binary volume the crossing is always at the edge midpoint, so only a finite set of triangle orientations can occur, and the mesh approximates a plane of arbitrary orientation by a small-scale zig-zag whose excess area depends on the orientation but not on the voxel size. Local polyhedrization schemes of this kind are not multigrid convergent: their area error does not go to zero as the grid is refined [2, 3]. On clean spheres we measure +7.3% at 16^3, rising to +8.8% at 128^3.

On a greyscale volume the vertex position is linearly interpolated from the partial-volume intensities, which recovers sub-voxel position information. For a plane and a symmetric blur the 0.5 iso-surface of the blurred image coincides with the true surface. For a curved surface it does not. If the effective blur is Gaussian with variance σ_{eff}^2, the level sets of the blurred indicator move towards the centre of curvature by

$$\delta \approx \sigma_{\text{eff}}^2 H, \qquad H = \tfrac{1}{2}(\kappa_1 + \kappa_2), \tag{2}$$

which follows from the heat equation: at the 0.5 level of a blurred step the second normal derivative vanishes, so the normal speed of the level set equals the sum of principal curvatures, applied over a diffusion time $\sigma_{\text{eff}}^2/2$. For a sphere of radius R this gives a relative area error of $-2\delta/R = -2\sigma_{\text{eff}}^2/R^2$. In our forward model the voxel box ($\sigma^2 = 1/12$) and the Gaussian PSF ($\sigma = 0.6$) combine to $\sigma_{\text{eff}}^2 \approx 0.443$ voxel^2; adding a smoothing filter with $\sigma_s = 1$ raises it to 1.443. Table 1 compares this prediction with the measured errors on clean spheres. The agreement is close at low resolution, where the curvature term dominates; at 64^3 the remaining discrepancy is of the same size as the interpolation error of marching cubes itself.

3.3 Effect of noise

Additive noise with standard deviation σ_n has two effects on marching cubes. First, it perturbs the iso-crossing positions near the surface. A randomly perturbed surface has more area than the smooth one, since the area element $\sqrt{1+|\nabla\eta|^2} \geq 1$ for any height perturbation η, so this bias is always positive. Second, background voxels exceed the iso-level by chance with probability $p = Q(0.5/\sigma_n)$, where Q is the Gaussian tail function, and object voxels fall below it with the same probability. Each such voxel becomes a tiny closed surface or cavity. For $\sigma_n = 0.15$, $p = Q(3.33) \approx 4.3 \times 10^{-4}$, which predicts about 112 spurious components on a 64^3 grid and 14 on a 32^3 grid. Each component adds one to the Euler characteristic of the thresholded volume, and the measured Euler numbers at this noise level are indeed around 100 to 125 at 64^3. For $\sigma_n = 0.05$, $p = Q(10)$ is negligible and only the surface roughening remains. Gaussian smoothing removes most of this noise, but by Eq. (2) it also shrinks curved surfaces, and it erases structures thinner than about $2\sigma_s$ entirely (we observe this for the thinnest tori on the 32^3 grid, where the smoothed volume never reaches the iso-level).

The upshot is that greyscale marching cubes has a negative, curvature-driven bias that grows at low resolution, plus a positive, noise-driven bias that grows with σ_n and with the number of voxels. The two can partly cancel, which is why a single correction factor cannot fix both.

3.4 Reference surface area

Spheres, cylinders and tori have closed-form areas. For an ellipsoid with semi-axes $a \geq b \geq c$,

$$A = 2\pi c^2 + \frac{2\pi ab}{\sin\varphi}\left[E(\varphi, m)\sin^2\varphi + F(\varphi, m)\cos^2\varphi\right], \quad \cos\varphi = \frac{c}{a}, \quad m = \frac{a^2(b^2-c^2)}{b^2(a^2-c^2)}, \tag{3}$$

with F and E the incomplete elliptic integrals of the first and second kind. The remaining families (random blobs and superellipsoids) are star-shaped, i.e. their surface is $\{r(\mathbf{u})\,\mathbf{u} : |\mathbf{u}| = 1\}$ for a radial function r. We triangulate them by mapping the vertices of a subdivided icosahedron to radius $r(\mathbf{u})$. For a smooth surface the area of such an inscribed mesh converges as $A_\ell = A + c\,h_\ell^2 + o(h_\ell^2)$ in the edge length h_ℓ, and each subdivision halves h_ℓ, so Richardson extrapolation of two levels,

$$A_{\text{ref}} = \tfrac{1}{3}\left(4A_{\ell+1} - A_\ell\right), \tag{4}$$

cancels the leading error term. We use levels 6 and 7 (81,920 and 327,680 triangles) and keep $|A_{\text{ref}} - A_7|$ as an error estimate for each shape.

3.5 Learning target

All models predict

$$y = \log\frac{A_{\text{true}}}{A_{\text{MC}}}, \tag{5}$$

where A_{MC} is the greyscale marching-cubes area, and the final estimate is $\hat{A} = A_{\text{MC}}\, e^{\hat{y}}$. The target is dimensionless, so it does not depend on the size of the object, and it is invariant to rotations and translations of the object. This lets the training data be augmented exactly with the 48 symmetries of the cube (axis permutations and reflections), which permute voxels without interpolation and leave the true area unchanged.

4 Previous work

Mesh-based measurement. Marching cubes [1] is the standard way to extract an iso-surface from a voxel volume; we use the topologically consistent variant of Lewiner et al. [4] as implemented in scikit-image [5]. Nyström et al. [6] compared areas of digital (voxel-face) surfaces and triangulated surfaces on phantoms and real objects and found the triangulated areas considerably more accurate, with errors decreasing as the voxel size decreased.

Local estimators and multigrid convergence. A large body of work in digital geometry estimates surface area from binary volumes by assigning a weight to each local $2 \times 2 \times 2$ configuration. Mullikin and Verbeek [7] derived optimal weights for digitized planes, and Windreich et al. [8] and Lindblad [9] optimised local configuration weights to minimise the bias for isotropically oriented surfaces. A central theoretical question is multigrid convergence, i.e. whether the estimate converges to the true area as the grid is refined. Kenmochi and Klette [2] showed that local polyhedrization schemes such as marching cubes on binary data are not multigrid convergent even for spheres, cubes and cylinders, and Svane [3] proved that no local digital algorithm for surface area is multigrid convergent on the class of convex polytopes. Ziegel and Kiderlen [10] studied design-based surface-area estimation from digitizations with random grid placement. Klette and Rosenfeld [11] give a general account. These results explain why the binary estimators in this work keep a resolution-independent bias, and they motivate using greyscale information.

Learning-based estimation. Deep networks operating directly on voxel grids [12] and on unordered point sets [13] are established tools for 3D shape analysis. For CT of porous materials, Rabbani et al. [14] trained a convolutional network to predict a range of morphological, hydraulic, electrical and mechanical properties of porous samples from 3D images in a fraction of the time of image-based simulation. Random Forests [15] on hand-crafted descriptors remain a strong baseline for small tabular problems.

Gap. The work above either derives fixed estimators for binary data or learns global properties of complex microstructures, where an exact ground truth is not available. We did not find a study that measures how much a learned model reduces surface-area bias relative to marching cubes against an exact reference, separately for resolution, noise level and unseen shape families, and that compares deep models with a calibrated feature-based baseline. That comparison is the subject of this work.

5 Method

5.1 Shape families

Every shape is defined in a body frame with maximum extent 1 by an implicit function that is negative inside. Five families are used for training and two are held out for testing only (Table 2). The random blobs are star-shaped with

$$\log r(\mathbf{u}) = \sum_k a_k \cos(\omega_k \, \mathbf{d}_k \cdot \mathbf{u} + \phi_k) \,, \tag{6}$$

a sum of plane waves on the sphere with random unit directions $\mathbf{d}_k$ and phases ϕ_k, which behaves like a Gaussian random field on the sphere. Smooth blobs use six low-frequency waves; rough blobs add ten high-frequency, low-amplitude waves. Superellipsoids $|x/a|^p + |y/b|^p + |z/c|^p = 1$ are drawn either with $p < 2$ (pointed tips) or $p > 2$ (box-like, with rounded edges of high curvature), so they test both kinds of geometry the training families do not contain.

5.2 CT-like voxelization

Each shape is scaled so that its maximum extent is a fraction $f \sim \mathcal{U}[0.35, 1]$ of the half grid (minus a 2-voxel margin), rotated by a uniformly random rotation, and offset by a random sub-voxel shift. Object diameters range from 9.8 to 60 voxels. The volume is then formed by a simple CT forward model:

1. *Partial-volume occupancy.* Each voxel value is the fraction of the voxel inside the shape, computed with $5 \times 5 \times 5$ supersampling in a band of ± 4 voxels around the surface. The summed occupancy matches the true volume to within 10^{-4} for every family.

Table 2: Shape families. Parameters are drawn uniformly from the given ranges; all shapes are normalised to maximum extent 1 before placement.

Family	Role	Shapes	Parameters
Sphere	train	150	radius 1
Ellipsoid	train	350	$a = 1$, $b, c \in [0.35, 1]$
Cylinder (capped)	train	300	radius $\in [0.3, 0.85]$, half-height $\sqrt{1-\rho^2}$
Torus	train	300	$r/R \in [0.2, 0.6]$, $R + r = 1$
Smooth blob	train	900	6 waves, $\omega \in [1, 3]$, s.d. of $\log r \in [0.15, 0.30]$
Superellipsoid	held out	200	$b, c \in [0.45, 1]$; $p \in [1.5, 1.8]$ (40%) or $[3, 6]$ (60%)
Rough blob	held out	200	smooth blob + 10 waves, $\omega \in [8, 14]$, s.d. $\in [0.02, 0.05]$

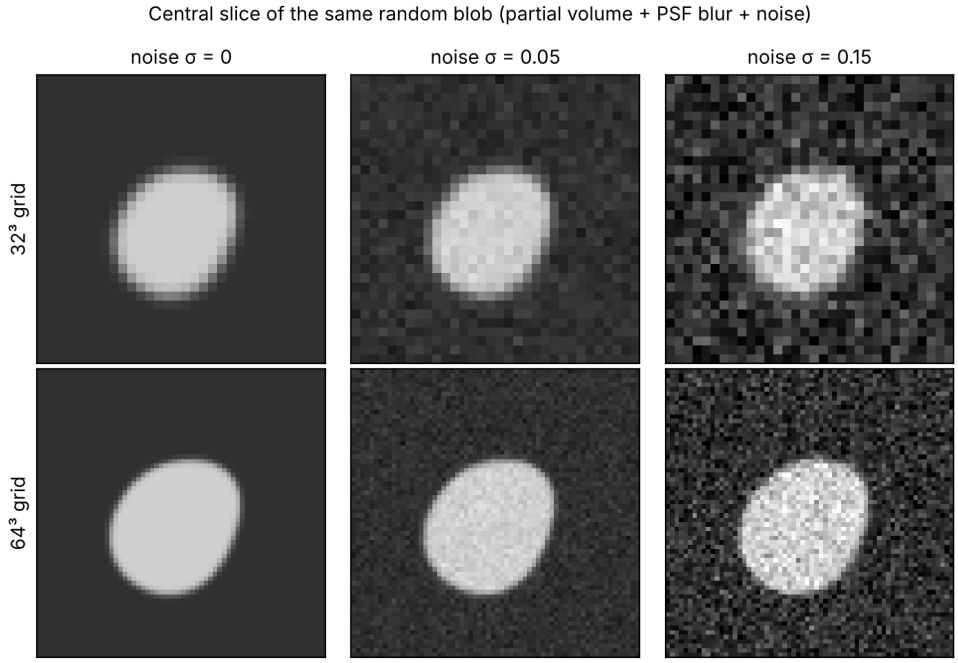


Figure 2: Central slice through the same random blob at both resolutions and all three noise levels.

2. *System blur.* A Gaussian point-spread function with $\sigma = 0.6$ voxel.
3. *Noise.* Additive Gaussian noise with $\sigma_n \in \{0, 0.05, 0.15\}$ relative to a unit object/background contrast.

Volumes are stored as 8-bit integers covering intensities $[-0.5, 1.5]$ (step 0.0078). Each shape is rendered at $N = 32$ and $N = 64$ with the same rotation and fill fraction, at all three noise levels, giving six volumes per shape and 14,400 volumes in total. Figure 2 shows an example.

The data are split by shape instance (70/15/15 within each training family), so all six variants of a shape fall into the same split and no information leaks between training and test. This gives 1,400 training, 299 validation and 301 test shapes from the training families, plus 200 superellipsoids and 200 rough blobs that are used only for testing.

5.3 Classical estimators

- **Voxel faces**: threshold at 0.5 and count exposed faces.
- **MC (binary)**: marching cubes on the thresholded volume.
- **MC (greyscale)**: marching cubes on the greyscale volume at iso-level 0.5.

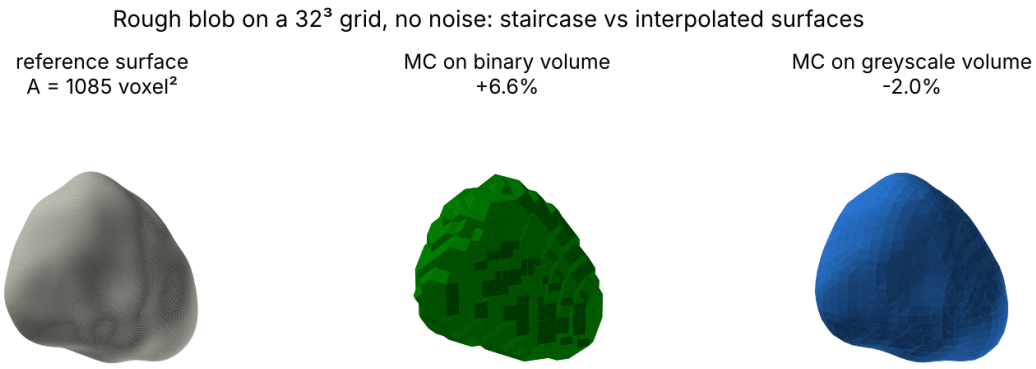


Figure 3: Reference surface of a rough blob and the marching-cubes meshes extracted from its clean 32^3 volume. The binary mesh shows the staircase pattern; the greyscale mesh is smooth but loses the finest surface detail.

- **MC (smoothed)**: Gaussian smoothing with $\sigma_s = 1$ voxel, then marching cubes at 0.5.

Figure 3 contrasts the binary and greyscale meshes.

5.4 Calibrated baselines

MC + lookup multiplies the greyscale MC area by $e^{\bar{y}}$, where $\bar{y}$ is the median training target for the same resolution and noise level. It is given the true noise level, which makes it optimistic; any learned model has to beat it to show that it does more than memorise one correction per regime. **Random Forest** [15] (400 trees, minimum leaf size 3, half of the features per split) predicts y from ten hand-crafted descriptors: $\log(A_{\mathrm{MC}}/N^2)$, the log ratios of the face, binary-MC and smoothed-MC areas to A_{MC}, a compactness term $\log(V^{2/3}/A_{\mathrm{MC}})$ with V the summed intensity, two principal-axis aspect ratios, the Euler number of the thresholded volume, a background noise estimate, and N.

5.5 Learned models

3D CNN. Five convolutional blocks with 16, 32, 64, 128 and 256 channels. The first block has one 3^3 convolution, the others two, each followed by batch normalisation and ReLU, and each block ends with $2\times$ max pooling. Global average and max pooling feed a head of 128 hidden units with dropout 0.2 and one output (3,593,297 parameters). The input intensity is rescaled to $[-1, 1]$.

PointNet. 2,048 points are sampled uniformly by area on the greyscale marching-cubes mesh. Coordinates are centred on the grid centre and divided by a fixed constant (not per sample), so the point cloud keeps the voxel scale. A shared MLP of widths (64, 64, 64, 128, 1024), global max pooling and a head of 512 and 128 units with dropout 0.3 follow Qi et al. [13] (742,209 parameters). Instead of the T-Net alignment module we apply a uniformly random rotation to every training sample, which is exact because the target is rotation invariant.

Hybrid CNN. The 3D CNN with the ten standardised Random Forest descriptors concatenated to the pooled image features before the head (3,594,577 parameters). It tests whether learned image features add information beyond the hand-crafted ones.

Table 3: Surface-area error (%) on held-out shapes, all noise levels pooled. MAPE: mean absolute error; bias: mean signed error. The two right-hand blocks are shape families never seen in training (MAPE). Best value per column in bold.

	Test MAPE		Test bias		Superellipsoid		Rough blob	
Method	32^3	64^3	32^3	64^3	32^3	64^3	32^3	64^3
Voxel faces	56.71	65.54	+56.64	+65.54	57.45	64.81	55.30	65.65
MC (binary)	10.38	14.49	+9.56	+14.49	10.51	14.31	8.91	13.99
MC (greyscale)	4.78	3.52	−1.25	+2.41	4.10	3.38	5.27	3.57
MC (smoothed)	13.78	3.04	−13.78	−3.04	9.93	2.29	14.02	4.37
MC + lookup	2.97	0.79	−1.45	−0.46	1.87	0.43	3.22	1.09
Random Forest	0.72	**0.18**	−0.18	**+0.04**	**0.73**	0.39	**1.71**	**0.48**
PointNet	1.98	1.04	+0.33	+0.34	1.54	0.95	3.00	1.32
3D CNN	0.68	0.26	**−0.10**	+0.10	1.05	0.45	1.96	0.55
CNN + features	**0.62**	0.21	−0.15	+0.07	1.06	**0.39**	1.97	0.52

Training. One model per resolution. Huber loss (threshold 1) on $y/0.05$, so that the loss is quadratic for errors below 5%; AdamW with weight decay 10^{-4}; one-cycle learning-rate schedule [16] with peak 10^{-3}; batch size 32; up to 40 epochs (15 for the two CNN variants at 64^3, which cost about eight times more per epoch); early stopping after 10 epochs without improvement of the validation error, keeping the best checkpoint. The CNNs are augmented with a random cube symmetry per batch. Training used PyTorch [17] with the Metal (MPS) backend on an Apple-silicon laptop; one epoch of the 32^3 CNN takes about 60 seconds.

5.6 Evaluation

For each estimate $\hat{A}$ we compute the relative error $e = \hat{A}/A_{\text{true}} - 1$ and report the mean absolute percentage error (MAPE), the bias (mean signed error), its standard deviation, the 95th percentile of $|e|$, and R^2 of $\hat{A}$ against A_{true}. All numbers are on held-out shapes: the test split of the training families and the two held-out families.

6 Results

All results are on shapes that were not used for training or model selection: the test split of the five training families (301 shapes, 903 volumes per resolution) and the two held-out families (200 shapes each, 600 volumes per resolution). Table 3 gives the overall picture.

6.1 Classical estimators

Voxel-face counting overestimates the area by 57% at 32^3 and 66% at 64^3. This is more than the 50% predicted for clean, smooth surfaces because noise at $\sigma_n = 0.15$ adds isolated voxels whose faces also count; at that noise level the overestimate reaches 80% at 32^3 and 98% at 64^3 (Table 4).

Marching cubes on the binary volume has a positive bias (+9.6% at 32^3, +14.5% at 64^3) that grows with resolution instead of shrinking. The resolution sweep (Fig. 4) shows the same trend from 24^3 to 128^3: on clean data the bias rises from +0.5% to +8.5%, in line with the non-convergence results of Section 3; at $\sigma_n = 0.15$ it rises from +16% to +34%, because the number of spurious noise components grows with the number of voxels.

Greyscale marching cubes behaves as Section 3 predicts. Its bias is negative for small objects and low resolution (curvature shrinkage: −22.9% at 16^3 and −4.2% at 32^3 on clean data in the sweep) and becomes positive at high noise (+8.6% at 64^3 and +9.7% at 128^3 for $\sigma_n = 0.15$). Because the two effects have opposite signs, its error is not smallest on clean data: at 64^3 its test MAPE is 1.13% on clean data, 0.86% at $\sigma_n = 0.05$ and 8.56% at $\sigma_n = 0.15$. Smoothing before

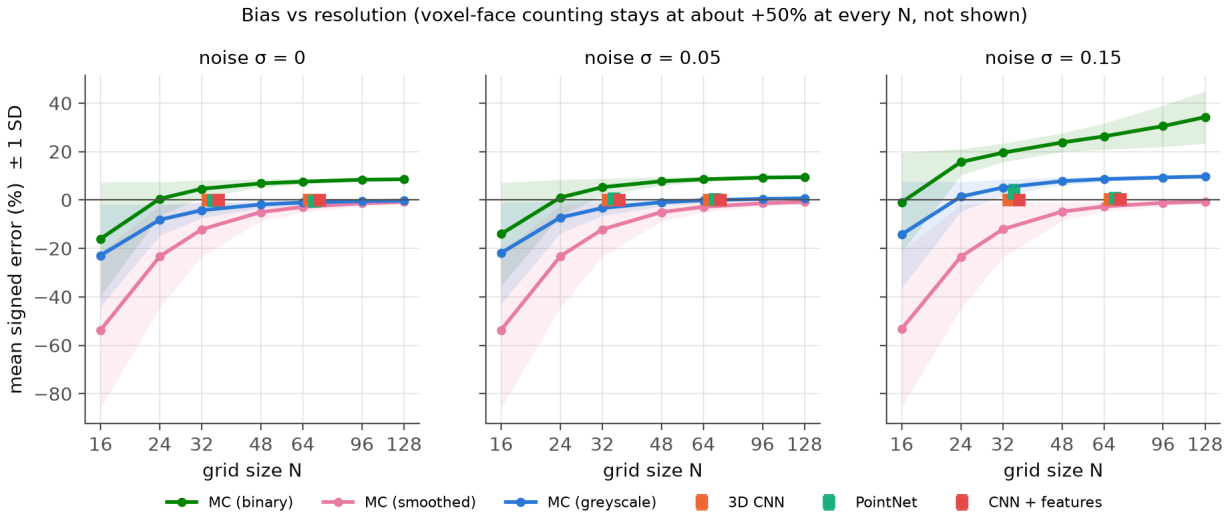


Figure 4: Mean signed error (±1 s.d.) of the marching-cubes estimators against grid size for 48 test shapes (8 per training family plus 8 superellipsoids), rendered with the same orientation and fill fraction at every N. Learned models are shown at the two sizes they were trained on.

marching cubes removes most of the noise bias (2.93% MAPE at 64^3, $\sigma_n = 0.15$), but it costs 13.8% at 32^3, where most objects are small enough for the curvature shrinkage to dominate, and it erases thin tori completely (29.7% MAPE on tori at 32^3, Table 6).

6.2 Learned and calibrated corrections

Every correction model improves on raw greyscale marching cubes by a large factor (Table 3, Fig. 5). The simple lookup table, one factor per resolution and noise level, already reduces the test MAPE from 4.78% to 2.97% at 32^3 and from 3.52% to 0.79% at 64^3. That it works much better at 64^3 shows that most of the remaining error at 64^3 is a systematic, regime-dependent bias, while at 32^3 the error depends strongly on the individual object.

The 3D CNN reaches 0.68% at 32^3 and 0.26% at 64^3, a seven-fold and a thirteen-fold reduction relative to greyscale marching cubes, with a bias within ±0.1% and R^2 of 0.99985 and 0.99997. The Random Forest on hand-crafted descriptors performs almost identically at 32^3 (0.72%) and is better at 64^3 (0.18%). PointNet is clearly weaker than both (1.98% and 1.04%): sampling 2,048 points from a mesh that already contains the noise artefacts throws away the intensity information that the volume-based models use.

The hybrid network, which receives the ten Random Forest descriptors next to its image features, gives the lowest error at 32^3 (0.62%) and the second lowest at 64^3 (0.21%, against 0.18% for the Random Forest). Its gain over the image-only CNN is small (0.06 and 0.05 percentage points), and it does not generalise better: on the held-out families it matches the CNN at 32^3 (1.06% and 1.97%) and the Random Forest at 64^3 (0.39% and 0.52%). Learned image features therefore add a little information beyond the physics-motivated descriptors on familiar shapes, but most of the correction is already captured by the descriptors themselves.

The Random Forest's feature importances explain why a small model does so well. The most important descriptors (impurity-based importance in brackets) are the Euler number (0.29), the noise estimate (0.26), the smoothed-to-raw marching-cubes area ratio (0.22) and the face-count-to-marching-cubes area ratio (0.08). The Euler number and the noise estimate measure exactly the two noise effects of Section 3 (spurious components and surface roughening), and the ratio of smoothed to raw marching-cubes area measures how much of the surface responds to smoothing, which is a proxy for curvature and thin features.

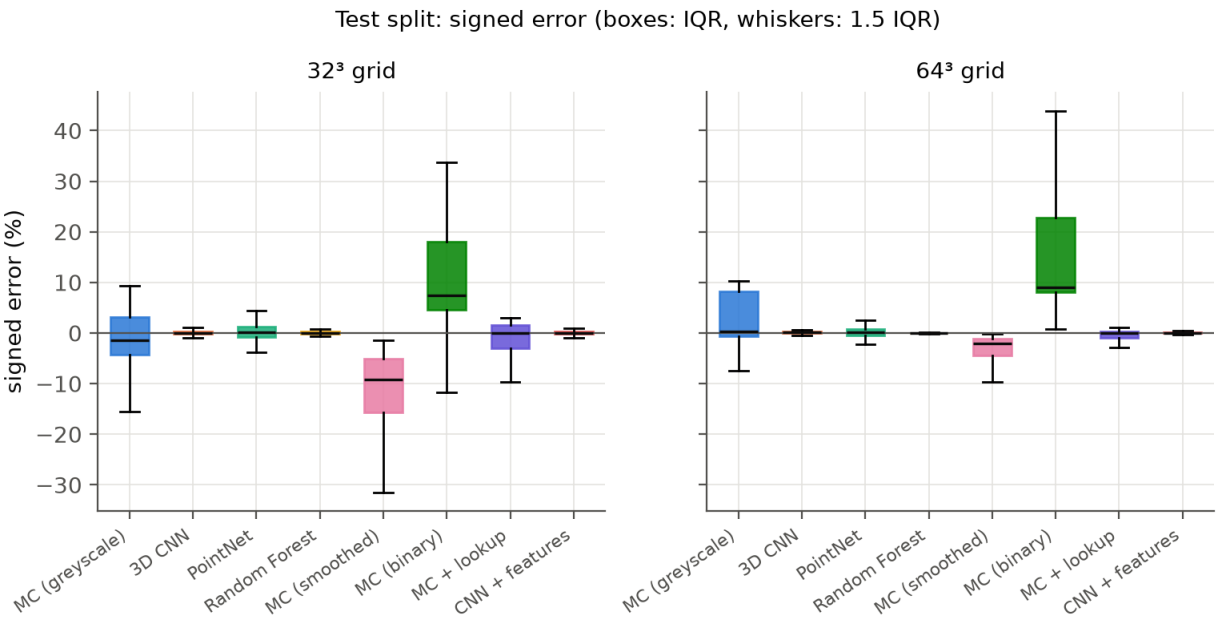


Figure 5: Distribution of the signed error on the test split. Boxes: interquartile range; whiskers: 1.5 IQR. Voxel-face counting is omitted for scale.

Table 4: Test MAPE (%) by resolution and noise level. Best per column in bold.

	32^3			64^3		
Method	σ_n=0	0.05	0.15	0	0.05	0.15
Voxel faces	44.10	45.64	80.38	48.31	49.91	98.40
MC (binary)	5.24	5.85	20.05	7.64	8.49	27.34
MC (greyscale)	4.73	3.84	5.77	1.13	0.86	8.56
MC (smoothed)	13.87	13.82	13.65	3.11	3.09	2.93
MC + lookup	3.06	3.05	2.80	0.84	0.83	0.71
Random Forest	0.49	0.61	1.07	**0.09**	0.14	**0.30**
PointNet	1.42	1.41	3.13	0.75	0.69	1.67
3D CNN	0.46	0.55	1.02	0.20	0.19	0.40
CNN + features	**0.44**	**0.49**	**0.93**	0.11	**0.14**	0.37

A failure case. The error distributions are heavy-tailed at 32^3: the standard deviation of the CNN error is 3.1% although its MAPE is 0.68%. Most of this comes from one test torus whose tube is 1.77 voxels thick. After the point-spread blur most of the tube stays below the iso-level, so greyscale marching cubes recovers only 42 of its 142 voxel^2 (and the smoothed volume nothing at all). All correction models fail on it, with errors between −47% and −66% across models and noise levels. A multiplicative correction of the marching-cubes area cannot recover a surface that is mostly missing from the mesh; structures thinner than about two voxels need a different approach.

6.3 Noise and object size

Table 4 and Fig. 6 separate the noise levels. The learned models are nearly insensitive to noise: at 64^3 the CNN's MAPE goes from 0.20% (clean) to 0.40% ($\sigma_n = 0.15$), while greyscale marching cubes goes from 1.13% to 8.56%. The same holds at 32^3 (0.46% to 1.02%, against 4.73% to 5.77%).

Object size matters more than grid size (Table 5). For objects less than 16 voxels across, greyscale marching cubes has a MAPE of 6.43%, the Random Forest 1.41%, the CNN 1.24% and the hybrid 1.14%; this is the size class in which the image-based models have the clearest advantage. For objects larger than 32 voxels the Random Forest is best, reaching 0.08% above 48

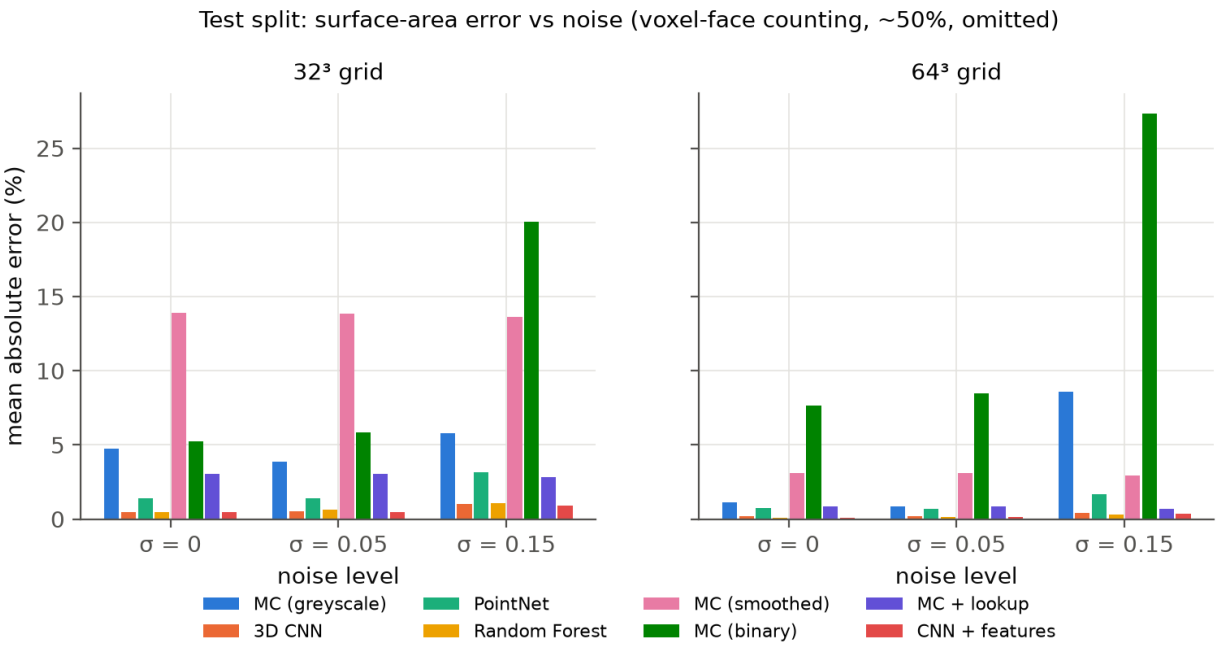


Figure 6: Test MAPE against noise level at both resolutions. Voxel-face counting (44–98%) is omitted.

Table 5: Test MAPE (%) by object diameter in voxels (both resolutions pooled).

Method	< 16	16–24	24–32	32–48	48–64
MC (greyscale)	6.43	3.89	3.57	3.45	3.44
MC (smoothed)	24.52	8.08	4.84	2.53	1.45
MC + lookup	4.76	1.78	1.48	0.67	0.58
Random Forest	1.41	0.39	**0.22**	**0.13**	**0.08**
PointNet	2.14	1.57	1.54	0.81	1.48
3D CNN	1.24	0.39	0.28	0.22	0.22
CNN + features	**1.14**	**0.35**	0.24	0.17	0.17

voxels, and even the lookup table stays below 0.7%.

6.4 Unseen shape families

Both held-out families stay far below the classical errors (Fig. 7, Table 3). On superellipsoids, whose sharp edges and pointed tips do not occur in the training data, the CNN reaches 1.05% and 0.45% and the Random Forest 0.73% and 0.39%, compared with 4.10% and 3.38% for greyscale marching cubes. These errors are between 1.0 and 2.2 times the test-split errors of the same models, so there is a measurable but moderate generalisation gap.

Rough blobs are the hardest case for every method: 1.96% (CNN) and 1.71% (Random Forest) at 32^3. Here the corrections are biased low (mean signed error -1.8% and -1.5%), as expected: the training shapes are smooth, so the models have never learned to add back area hidden in sub-voxel roughness. At 64^3, where more of the roughness is resolved, the error falls to about 0.5% and the bias vanishes. Table 6 breaks the results down by family.

6.5 Training behaviour

Figure 8 shows the validation error during training. The 32^3 CNN had nearly converged after 40 epochs. Both 64^3 CNN variants, limited to 15 epochs by the compute budget, reached their best validation error in the last epoch, so their results are likely not the best these architectures can reach. PointNet's validation error flattens early at both resolutions.

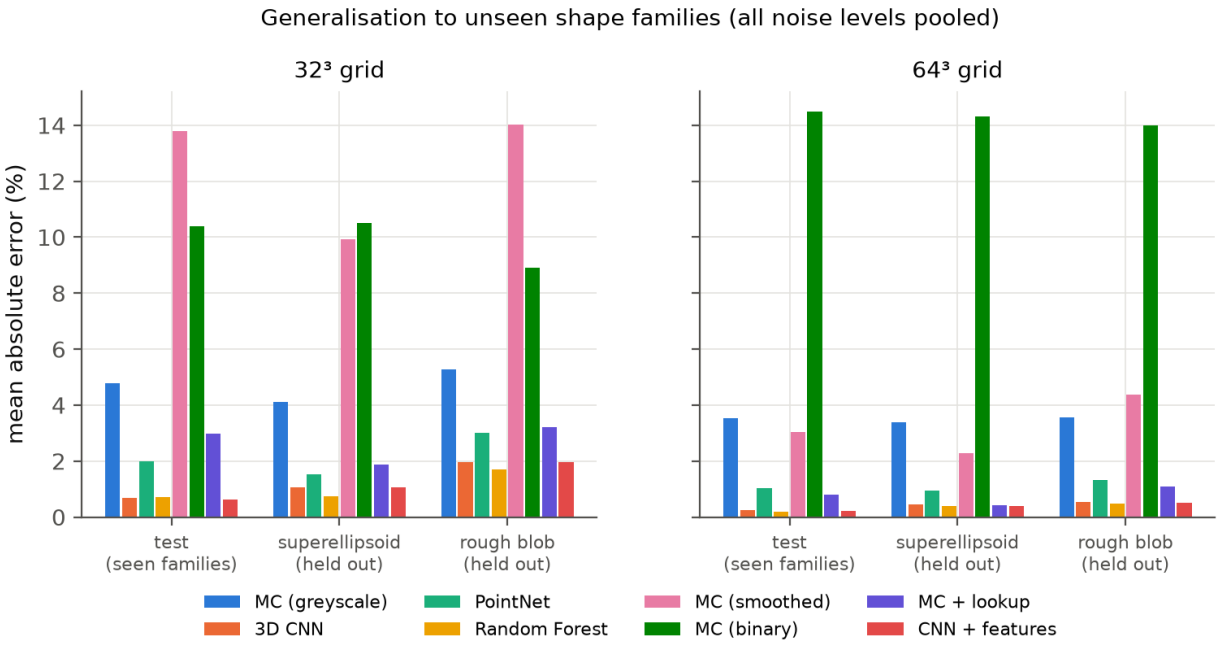


Figure 7: MAPE on the test split and on the two held-out families (all noise levels pooled).

Table 6: MAPE (%) by shape family (test split and held-out families, all noise levels). †Held out from training. Best per row and resolution in bold.

	32^3						64^3					
Family	MC grey	Lookup	RF	PointNet	3D CNN	Hybrid	MC grey	Lookup	RF	PointNet	3D CNN	Hybrid
Sphere	3.46	1.98	**0.19**	2.21	0.23	0.22	3.41	0.39	**0.06**	1.14	0.13	0.14
Ellipsoid	4.22	2.36	0.52	2.00	0.45	**0.44**	3.45	0.48	**0.20**	1.20	0.28	0.22
Cylinder	5.99	4.75	1.15	2.60	0.78	**0.64**	3.93	2.42	0.31	1.58	0.37	**0.30**
Torus	6.89	5.26	1.90	2.83	1.85	**1.79**	3.53	0.87	0.26	1.00	0.28	**0.22**
Smooth blob	4.12	2.03	**0.35**	1.45	0.41	0.36	3.43	0.41	**0.11**	0.79	0.23	0.18
Superellipsoid†	4.10	1.87	**0.73**	1.54	1.05	1.06	3.38	0.43	0.39	0.95	0.45	**0.39**
Rough blob†	5.27	3.22	**1.71**	3.00	1.96	1.97	3.57	1.09	**0.48**	1.32	0.55	0.52

6.6 Which estimator to use

Table 7 summarises when a learned correction is worth the effort. Its benefit is largest where the classical methods are weakest: small objects, coarse grids and noisy data. For large, clean objects greyscale marching cubes is already accurate to about 1%, and the gain from a learned model, while real, matters only when sub-percent accuracy is needed.

7 Conclusion

The surface-area problem stated in Section 2 is largely solved within the setting studied here. The errors of the classical estimators are not random: voxel-face counting is fixed at about +50% by geometry, binary marching cubes keeps a resolution-independent bias, and greyscale marching cubes combines a curvature shrinkage that a heat-equation argument predicts closely with a positive noise bias that grows with the number of voxels. Because these two effects depend on the object and on the acquisition, no single correction factor removes them.

A correction learned on synthetic data with exact ground truth does remove most of them. On unseen test shapes it reduces the error of greyscale marching cubes from 4.78% to 0.62% at 32^3 and from 3.52% to 0.18% at 64^3, it is nearly insensitive to noise, and it transfers to shape families with sharp edges or rough surfaces that were never in the training data, with errors below 2%.

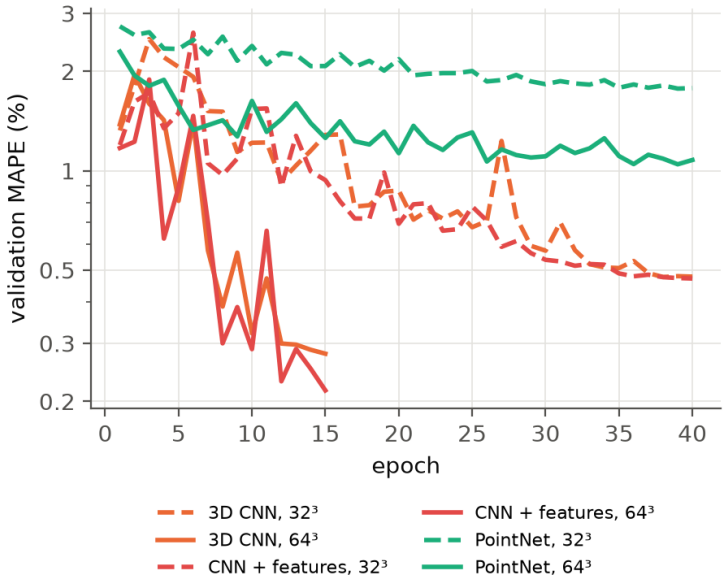


Figure 8: Validation MAPE during training (log scale).

Table 7: Recommended estimator by regime, based on the test results.

Regime	Recommendation
Any noise level, any size	Do not use voxel-face counting or binary marching cubes for absolute area; their bias is large and does not decrease with resolution.
Clean data, large objects	Greyscale marching cubes is within about 1% at 64^3; a per-regime correction factor brings it below 1%.
Noisy data ($\sigma_n = 0.15$)	Raw marching cubes overestimates by up to 9–10%. Smoothing helps only for large objects; a learned correction (CNN or Random Forest) gives 0.3–1.1%.
Small objects (< 16 voxels)	Use an image-based model (hybrid 1.14%, CNN 1.24%, against 6.43% for marching cubes); smoothing is harmful.
Structures thinner than $\sim$2 voxels	No method tested here is reliable.

The comparison between models is the most useful practical result. A Random Forest on ten physics-motivated descriptors, chief among them the Euler number and a noise estimate, matches a 3.6-million-parameter 3D CNN overall and is the best model for large objects; adding learned image features to those descriptors helps only slightly. Deep models earn their cost mainly for small objects and coarse grids, and a point-cloud model built on the noisy marching-cubes mesh is clearly inferior to models that see the volume. Two regimes remain unsolved: structures thinner than about two voxels, which marching cubes loses before any correction can act, and sub-voxel roughness, where the information is no longer present in the image.

The natural next steps are a projection-domain CT simulation with Poisson noise and reconstruction artefacts, volumes with many touching structures such as porous media and lattices, and validation on a real scan of an object with a known surface area.

8 Limitations

- **Simplified physics.** The forward model is partial-volume averaging, a Gaussian PSF and white Gaussian noise added in the image domain. Real CT noise is correlated by the reconstruction filter and is signal dependent, and real volumes contain beam-hardening, ring and

streak artefacts. A projection-domain simulation (forward projection, Poisson noise, filtered back-projection) would be the natural next step.

- **One isolated object per volume.** Porous media and lattices consist of many connected or touching structures, and the correction may depend on how much of the surface is close to other surfaces. The models have not been tested on such data.
- **Fixed acquisition parameters.** Blur, contrast and the iso-level are fixed, and the models are trained at two grid sizes and three noise levels. Their behaviour between and outside these settings is untested; in particular, a model trained at $\sigma_n \leq 0.15$ should not be trusted at higher noise.
- **Sub-voxel roughness is not observable.** Surface detail much smaller than a voxel does not survive voxelization. A learned model can only add back the excess area it expects from the roughness statistics of its training data, so its correction on rough surfaces is a prior, not a measurement, and it will be wrong if the real roughness spectrum differs.
- **Limited shape space.** The training shapes are star-shaped or of genus at most one. Shapes with many handles, thin sheets or concave pockets are not represented.
- **No tuning, single seed.** Architectures and hyperparameters were fixed in advance, the 64^3 CNNs were trained for only 15 epochs because of the compute budget (a single laptop), and each model was trained once. Differences of a few hundredths of a percent between models should not be over-interpreted.
- **Real data.** No measurement on a real CT scan with a known surface area is included.

Code and data availability

All code, trained weights and result tables are in the project repository. The dataset is fully determined by the generation script and its fixed seed (51) and can be regenerated with one command; the generated volumes are also shared alongside the code.

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